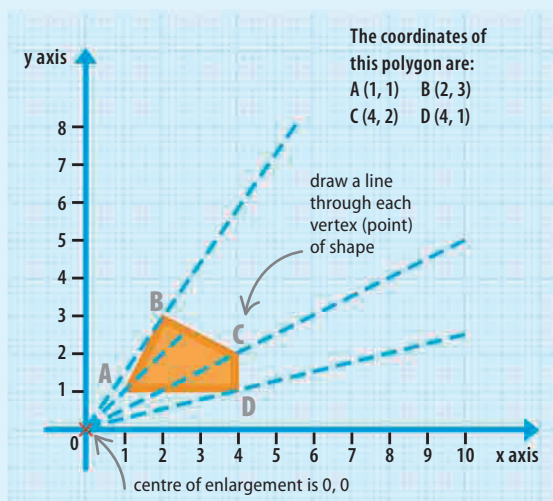
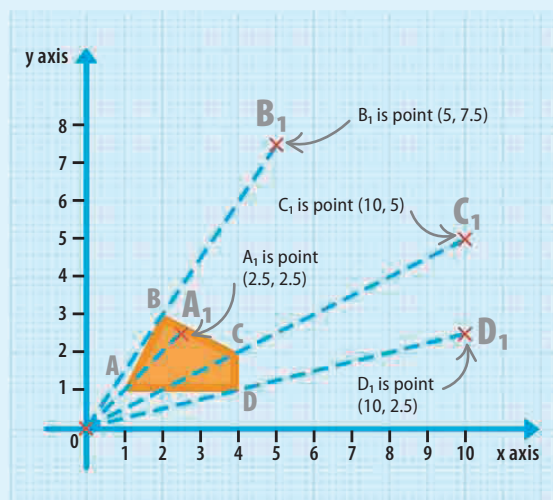


Constructing an enlargement

An enlargement is constructed by plotting the coordinates of the object on squared (or graph) paper. Here, the quadrilateral ABCD is measured through the centre of enlargement (0, 0) with a given scale factor of 2.5.



Draw the polygon ABCD using the given coordinates. Mark the centre of enlargement and draw lines from this point through each of the vertices of the shape (points where sides meet).



Read along the x axis and the y axis to plot the vertices (points) of the enlarged image. For example, B₁ is point (5, 7.5) and C₁ is point (10, 5). Mark and label all the points A₁, B₁, C₁, and D₁.

horizontal distance of A from centre of enlargement vertical distance of A from centre of enlargement x coordinate

$$A_1 = 1 \times 2.5, 1 \times 2.5 = (2.5, 2.5)$$

scale factor y coordinate

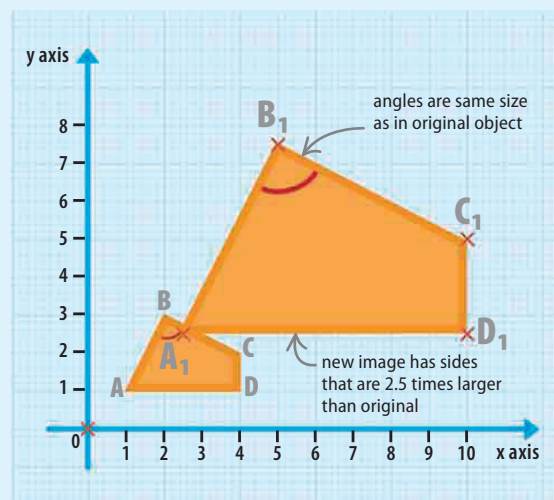
The same principle is then applied to the other points, to work out their x and y coordinates.

$$B_1 = 2 \times 2.5, 3 \times 2.5 = (5, 7.5)$$

$$C_1 = 4 \times 2.5, 2 \times 2.5 = (10, 5)$$

$$D_1 = 4 \times 2.5, 1 \times 2.5 = (10, 2.5)$$

Then, calculate the positions of A₁, B₁, C₁, and D₁ by multiplying the horizontal and vertical distances of each point from the centre of enlargement (0, 0) by the scale factor 2.5.



Join the new coordinates to complete the enlargement. The enlarged image is a quadrilateral with sides that are 2.5 times larger than the original object, but with angles of exactly the same size.